

FINAL REPORT
TOPIC: PARTICIPATION IN A KAGGLE
COMPETITION TO SOLVE A MACHINE
LEARNING PROBLEM

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OUTLINE



**CONTEXT
AND OBJECTIF**



**TECHNICAL
REPORT**



**ECONOMIC
REPORT**



CONCLUSION



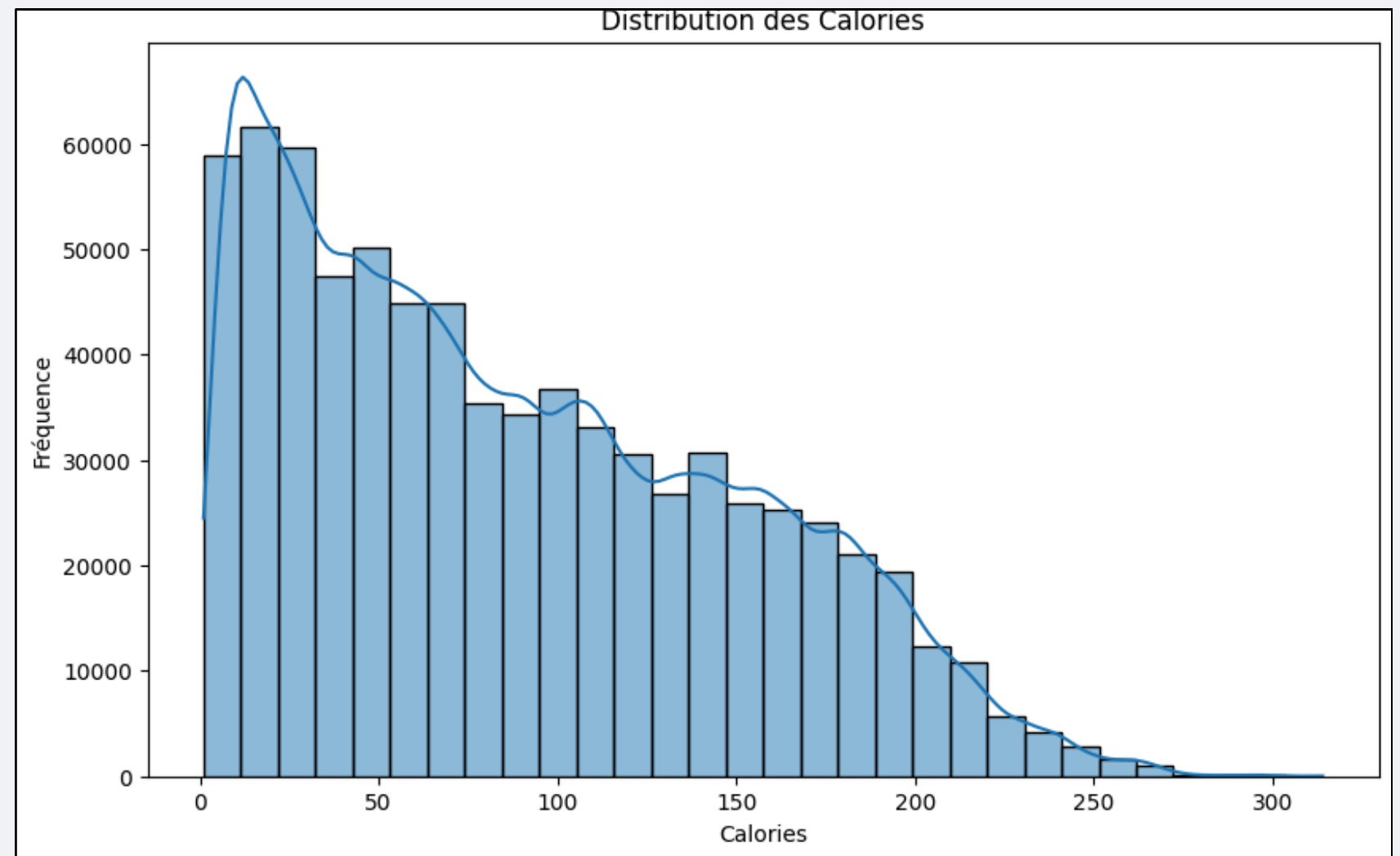
CONTEXT AND OBJECTIF

BASED ON NUMERICAL FEATURES SUCH AS HEIGHT (IN CM), WEIGHT (IN KG), AND ONE CATEGORICAL FEATURE — SEX — THE OBJECTIVE OF THIS KAGGLE COMPETITION IS TO PREDICT THE NUMERICAL TARGET VARIABLE “CALORIES” BURNED DURING A WORKOUT.

ECONOMIC REPORT

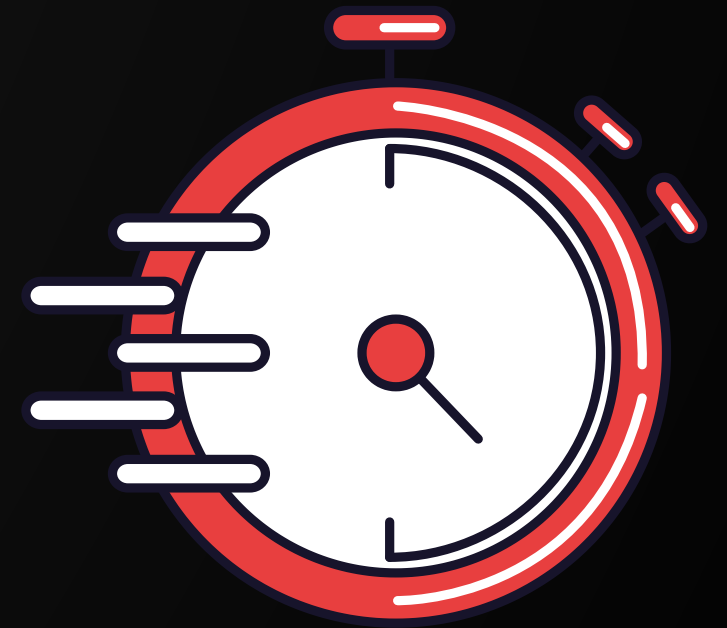
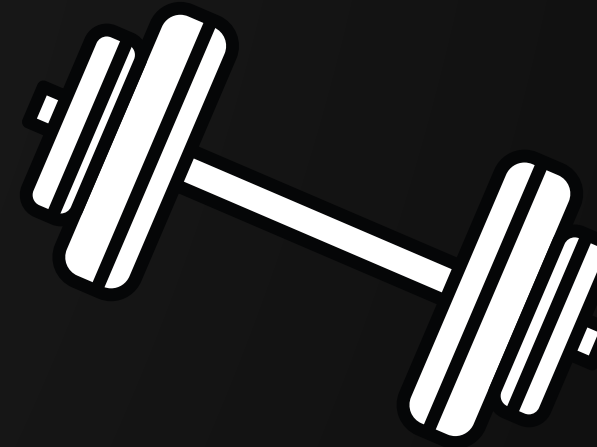
Dataset Summary

- Demographics: 750k participants aged 20–80 (peak at 20–22); 50.1% female / 49.9% male.
- Metrics: 1–30 min sessions; 140–220 cm height; 36–132 kg weight (36 severe BMI cases); 67–128 bpm heart rate; 37–41.5°C temperature.
- Target (Calories): Right-skewed, ranging from 1 to 314 cal.
- Outliers: All valid biological extremes (139 high-calorie, 36 high-HR, 14 height, 9 weight); no cleanup needed.
- Key Correlations: Calories correlate strongly with workout duration, heart rate, and temperature. All features display strong physiological consistency.



TECHNICAL REPORT

- The variable id was deleted because it is useless and could lead to data leakage.
- The model that achieved the highest performance was the Random Forest Classifier.
- For the feature importance, "duration" is the variable that contributes the most to predicting the target variable: "calories".



CONCLUSION

In summary, this study conducted on a dataset of 750,000 participants highlights highly consistent biological and physiological features. The exploratory data analysis confirmed that the distribution of variables—despite showing extreme values (outliers) for calories, heart rate, and physical build—remains medically plausible, allowing the entire sample to be retained without distorting the data.

From a modeling perspective, strategic data cleaning combined with the strong observed correlations proved effective. The Random Forest Classifier emerged as the top-performing model, primarily driven by workout duration, which remains the key predictor of calorie expenditure.



THANK YOU

